# A PEPS plugin for TNQVM

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## Summary

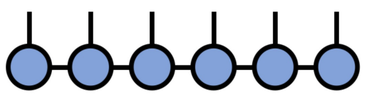



Fig (1) MPS [3]

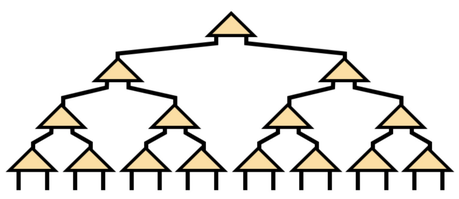


Fig (2) TTN [3]

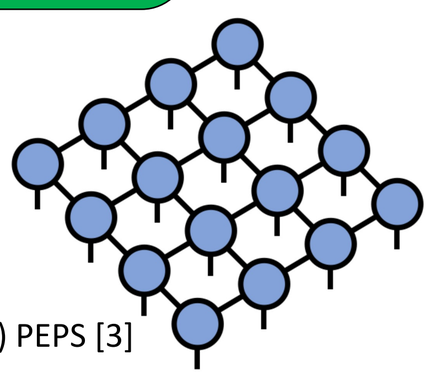


Fig (3) PEPS [3]

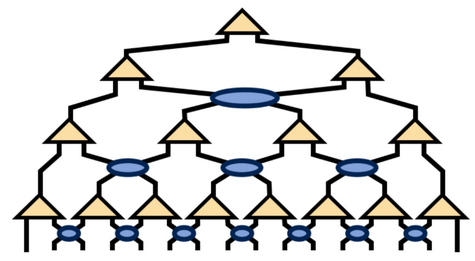


Fig (4) MERA [3]

**Here we integrate the Projected Entangled Pair State (PEPS) tensor network topology with Tensor Network Quantum Virtual Machine (TNQVM) and implement the snake boundary contraction algorithm leveraging pre-implemented topologies like MPS and MPO**

**2 Step Process** ➔ **Represent a Quantum Circuit as a PEPS tensor network**
➔ **Contract this tensor network using approximate algorithms for observables**


## Motivation

**2D Tensor Network (PEPS)**

- **Efficiently represent quantum many-body systems in two or higher dimensions**
- **2D Hamiltonians**

## Background

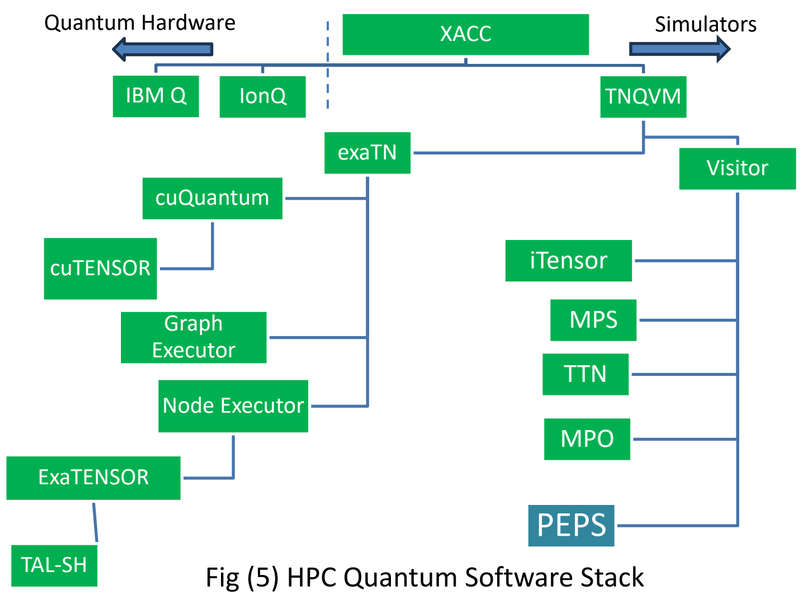



Fig (5) HPC Quantum Software Stack

## Approach

- **Snake Boundary Algorithm**
  - **Zig-Zag Pattern** [2]

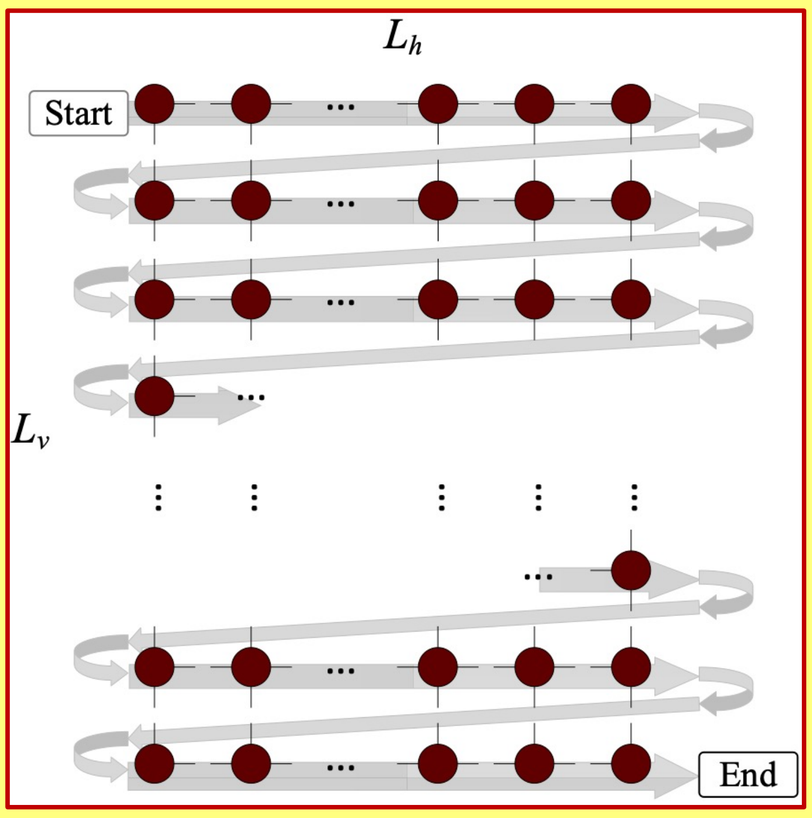



Fig (6) Snake PEPS contraction

- **Re-use existing visitor features**
- **Contraction sequence from above step → MPS Visitor**
- **Build a new MPS Tensor Network (ExaTN)**
- **Contract this MPS using pre-implemented methods**

## Quantum Circuit to TN

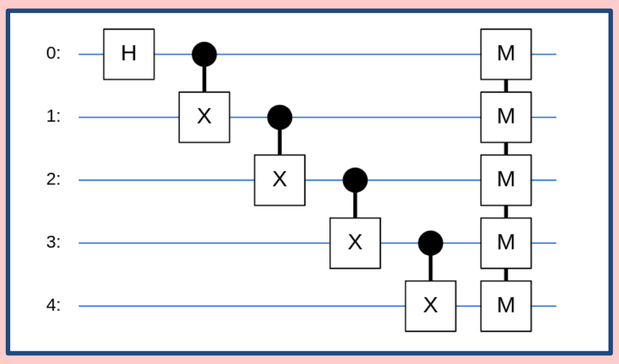



Fig (7) GHZ Quantum Circuit

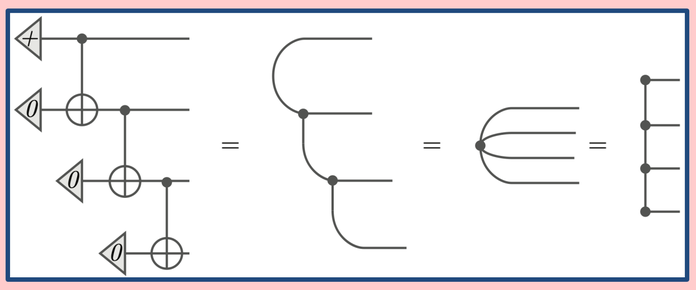


Fig (8) GHZ Quantum Circuit to MPS [4]

- **Conversion to PEPS can be tricky to visualize!**

**Algorithm 1** Conversion of Quantum Circuit to PEPS

```
1:  pepsNetwork ← exatn::builder("PEPS").initialize(X,Y);
2:  for gate_i in circuit.gates() do
3:      gateTensor ← tnqvm::getGateTensor(gate_i);
4:      if gate_i.rank() == 2 then
5:          mergedTensor ← pepsNetwork.getQubitTensor(i) ⊗
            gateTensor;
6:          pepsNetwork.setQubitTensor(a, mergedTensor);
7:      end if
8:      if gate_i.rank() == 4 then
9:          mergedTensor ← pepsNetwork.getQubitTensor(a) ⊗ pep-
            sNetwork.getQubitTensor(b);
10:         contractedTensor ← mergedTensor ⊗ gateTensor;
11:         U, V ← TruncatedSVD(contractedTensor);
12:         pepsNetwork.setQubitTensor(a, U);
13:         pepsNetwork.setQubitTensor(b, V);
14:     end if
15: end for
```

- **All methods used in the above algorithm are pre-implemented in TNQVM already**

## Plugin Usage (GHZ-9)

```
xacc::Initialize();
auto xasmCompiler = xacc::getCompiler("xasm");
auto ir = xasmCompiler->compile(R"(__qpu__
void test2(qbit q) {
   H(q[0]);
   for(int i = 0; i < 8; i++) {
      CNOT(q[i], q[i+1]);
   }
   for (int i = 0; i < 9; i++) {
      Measure(q[i]);
   }
}
)");
std::vector<int> bitstring(9, 0);
auto program = ir->getComposite("test2");
auto accelerator = xacc::getAccelerator(
"tnqvm", {
   std::make_pair(
      "tnqvm-visitor", "exatn-peps"
   ),
   std::make_pair("shots", 10),
   std::make_pair("lx", 3),
   std::make_pair("ly", 3)
});
auto qreg = xacc::qalloc(9);
accelerator->execute(qreg, program);
qreg->print();
```

GHZ Circuit

PEPS Visitor Meta

Execute

## Preliminary Results

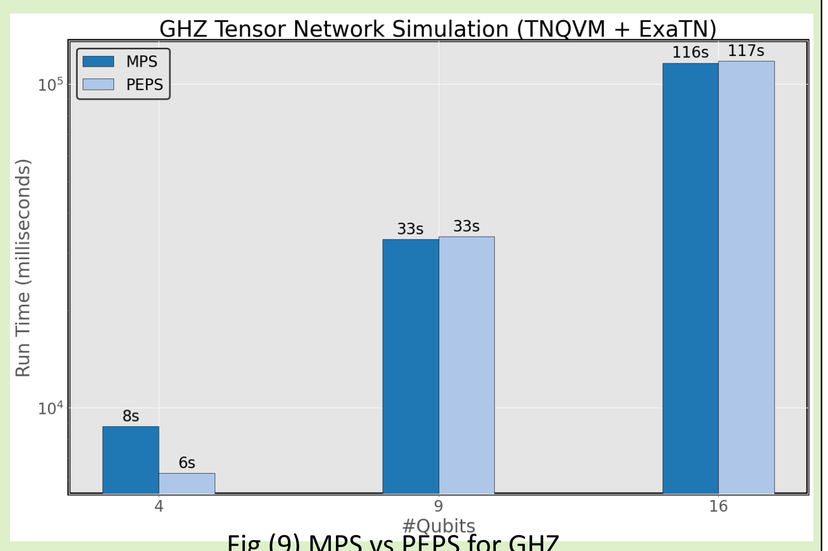



Fig (9) MPS vs PEPS for GHZ

- **Runtime(PEPS) ≈ Runtime(MPS)**
- **Test of correctness using the PEPS plugin for the GHZ quantum circuit**

## Conclusion

- **This new "visitor", PEPS, enables 2D tensor network simulation capturing multidimensional entanglement**
- **Merit: Utilizes pre-developed tensor network topologies, such as MPS and MPO → contraction of PEPS**

## Future Work

- **Extensive tests and benchmarking at scale, for problems asking for 2D entanglement representation**
- **Explore other PEPS approximate contraction algorithms**
- **Explore sparsity induced in PEPS Tensor Networks**

## References

[1] McCaskey, Alexander J. Tensor Network Quantum Virtual Machine (TNQVM). Computer software. https://www.osti.gov//servlets/purl/1340180. Vers. 00. USDOE. 18 Nov. 2016. Web.
[2] Guo C, Liu Y, Xiong M, et al. General-Purpose Quantum Circuit Simulator with Projected Entangled-Pair States and the Quantum Supremacy Frontier. Phys Rev Lett. 2019;123(19):190501. doi:10.1103/PhysRevLett.123.190501
[3] https://tensornetwork.org
[4] Jacob Biamonte and Ville Bergholm. "Tensor Networks in a Nutshell." 2017. arXiv:1708.00006 [quant-ph].

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